American Computer Science League

2024-2025 ● Contest 4: Short Problems Solutions ● Classroom Division

1. Graph Theory

The adjacency matrix with double arrows everywhere is as follows:

0	1	1	1	1	1
1	0	1	0	0	1
1	1	0	1	0	0
1	0	1	0	1	0
1	0	0	1	0	1
1	1	0	0	1	0

There are 20 entries of 1. Vertex A connects to all 5 other vertices and all of the other vertices connect to 3 others so 5+ 5*3 = 20.

C. 20

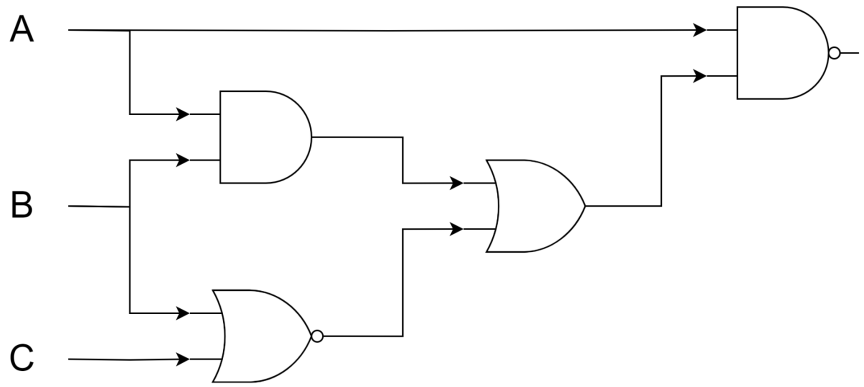
2. Graph Theory

By inspection, there are 6 of them: ABA, ABCA, ADCA, ADCBA, BCB, CDC.

D. 6

3. Digital Electronics

Translate the following digital circuit into a Boolean expression.



The gates from top to bottom are NAND, AND, OR, and NOR. The lower part is (A AND B) OR NOT (B OR C). That is $AB + \overline{B + C}$. Adding the top part gives you $\overline{A(AB + \overline{B + C})}$.

A. $\overline{A\,(AB + \overline{B + C})}$

4. Digital Electronics

This digital circuit is translated to $(A + B)(\overline{\overline{(B + C)}C}$. You could use a Truth Table, but simplifying the expression yields the following:

$(A + B)(\overline{\overline{(B + C)}C})$

$= (A + B)(\overline{\overline{B + C}} + \overline{C})$

$= (A + B)(B + C + \overline{C})$

$= (A + B)(1)$

$= A + B$

This expression is TRUE for 6 of the 8 possible ordered triples, all but (0,0,*).

E. 6

5. Graph Theory

The adjacency matrix raised to the 2nd power for paths of length 2 is as follows:

$$\begin{bmatrix} 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 \end{bmatrix}^2 = \begin{bmatrix} 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \end{bmatrix}$$

There are 8 entries of 1.

C. 8

6. Graph Theory

There are 11 cycles in the graph. They are: ABA, ADA, AEBA, AEDA, ABEDA, ADCEBA, ABCEDA, BEB, BCEB, CEC, CEDC.

D. 11

7. Digital Electronics

This circuit translates to the Boolean expression: $\overline{\overline{A(AB)} + \overline{B + C}}$

$$\begin{aligned} \overline{\overline{A(AB)} + \overline{B + C}} &= \overline{\overline{AB} + \overline{B + C}} \\ &= \overline{\overline{AB}}\,\overline{\overline{B + C}} \\ &= AB(B + C) \\ &= AB + ABC \\ &= AB(1 + C) \\ &= AB \end{aligned}$$

This is TRUE for (1,1,*). Therefore 2 triples make it TRUE

B. 2

8. Digital Electronics

This translates to $\overline{(A + B) \oplus BC + \overline{C}}$ since XOR is a higher priority than OR.

E. $\overline{(A + B) \oplus BC + \overline{C}}$

5. Assembly Language

```
NUM    DC      2025
       LOAD    NUM
       DIV     =100
       STORE   A
       MULT    =100
       SUB     NUM
       MULT    =-1
       STORE   B
       DIV     =5
       MULT    =10
       STORE   ANS
       LOAD    A
       DIV     =5
       ADD     ANS
       STORE   ANS
       PRINT   ANS
       END
```

This program contains only sequential statements that store the first 2 digits of 2025 in the variable A (20) and the last 2 digits of the 2025 in the variable B (25). ANS becomes 25/5 and multiplies it by 10 to make it the tens digit before adding 20/5 so the answer is 5*10+4=54, thus reversing the values of 25/5 and 20/5.

B. 54

6. Assembly Language

```
T      DC      0
N      DC      19
NUM    DC      135
TOP    LOAD    NUM
       SUB     T
       BE      STOP
       LOAD    N
       ADD     =1
       STORE   N
       LOAD    T
       ADD     N
       STORE   T
       BU      TOP
STOP   PRINT   N
       END
```

This program finds the sum of sequential numbers adding up to 135 and prints the last value that is needed to equal 135. The sum 20 + 21 + 22 + 23 + 24 + 25 = 135 so the answer is 25.

This problem was inspired by the fact that 179 + 180 + … + 189 = 2024.

E. 25